1) $\int_x^{x^2} (2x - y)\, dy = 2xy - \frac{y^2}{2}\Big|_x^{x^2} = 2x^3 - \frac{x^4}{2} - \frac{3x^2}{2} =$

$= 2x^3 - \frac{x^4}{2} - \frac{3x^2}{2}$

2) $\int_1^2 \left(2x^3 - \frac{x^4}{2} - \frac{3x^2}{2}\right) dx = \int \left(\frac{\cancel{2}x^4}{\cancel{4}_{2}} - \frac{x^5}{10} - \frac{\cancel{3}x^3}{\cancel{6}_{2}}\right) =$

$= \frac{x^4}{2} - \frac{x^5}{10} - \frac{x^3}{2} = \left(\frac{16}{2} - \frac{32}{10} - \frac{16}{2}\right) - \left(\frac{4}{2} - \frac{1}{10} - \frac{1}{2}\right) =$

$= \frac{4}{5} - \left(-\frac{1}{10}\right) = \frac{9}{10}$

2) $\int_{-1}^{1\ 2} dy \int_0^{y^2} (x \overset{1}{+} 2y)\, dx = \frac{1}{3}$

1) $\int_0^{y^2} (x + 2y)\, dx = \frac{x^2}{2} + 2yx\Big|_0^{y^2} = \frac{y^4}{2} + 2y^3$

2) $\int_{-1}^{1} \left(\frac{y^4}{2} + 2y^3\right) dy = \frac{y^5}{10} + \frac{\cancel{2}y^4}{\cancel{4}_{2}} = \frac{y^5 + 5y^4}{10}\Big|_{-1}^{1} =$

$= \frac{6}{10} - \frac{4}{10} = \frac{2}{10} = \frac{1}{5}$

g) $\int_4^6 \overset{2}{dx} \int_x^{2x} \frac{y}{x}^{1} dy = 15$

1) $\int_x^{2x} \frac{y}{x}\, dy = \frac{1}{x} \int_x^{2x} y\, dy = \frac{1}{x} \cdot \frac{y^2}{2}\Big|_x^{2x} = \frac{1}{x} \cdot \frac{3x^2}{2} = \frac{3x}{2}$

2) $\int_4^6 \frac{3x}{2}\, dx = \frac{3}{2} \cdot \frac{x^2}{2}\Big|_4^6 = \frac{3}{4}(36 - 16) = \frac{3}{4} \cdot 20 = 15$

Самостоятельная работа. Лиатова Ю.Р.
Вычисление двойных интегралов.

В 1

№1 $\int_{-1}^{1} dx \int_{1}^{2} (2x - y^2)\, dy = -\frac{14}{3}$

1) $\int_{1}^{2} (2x - y^2)\, dy = 2xy - \frac{y^3}{3}\Big|_1^2 = \left(2x \cdot 2 - \frac{8}{3}\right) - \left(2x - \frac{1}{3}\right) =$
$= \left(4x - \frac{8}{3}\right) - \left(2x - \frac{1}{3}\right) = 4x - \frac{8}{3} - \left(2x - \frac{1}{3}\right) =$
$4x - \frac{8}{3} - 2x + \frac{1}{3}) = 2x - \frac{7}{3}$

2) $\int_{-1}^{1} \left(2x - \frac{7}{3}\right) dx = \frac{2x^2}{2} - \frac{7}{3}\Big|_{-1}^{1} = 1 - 1 - \frac{7}{3} - (1 - (-1)) =$
$= -\frac{7}{3} \cdot 2 = -\frac{14}{3}$

№2 $\int_{2}^{3} dy \int_{0}^{1} (xy)\, dx = \frac{5}{4}$

1) $\int_{0}^{1} xy\, dx = y\, \frac{x^2}{2}\Big|_0^1 = y \cdot \frac{1}{2} = \frac{y}{2}$

2) $\int_{2}^{3} dy = \frac{1}{2} \cdot \frac{y^2}{2}\Big|_2^3 = \frac{1}{4}\,(9 - 4) = \frac{1}{4} \cdot 5 = \frac{5}{4}$

№3

$$\int_1^3 dx \int_{x^2}^{x} (x-y)\, dy =$$

1) $\int_{x^2}^{x} (x-y)\, dy = xy - \frac{y^2}{2} \Big|_{y=x^2}^{y=x} = \frac{x^2}{2} - \left(x^3 - \frac{x^4}{2}\right) =$

$\frac{x^2}{2} - x^3 + \frac{x^4}{2}$

2) $\int_1^3 \left(\frac{x^4}{2} - x^3 + \frac{x^2}{2}\right) dx = \frac{x^5}{10} - \frac{x^4}{4} + \frac{x^3}{6} =$

$= \left(\frac{3^5}{10} - \frac{3^4}{4} + \frac{3^3}{6}\right) - \left(\frac{1^5}{10} - \frac{1^4}{4} + \frac{1^3}{6}\right) = \frac{128}{15}$